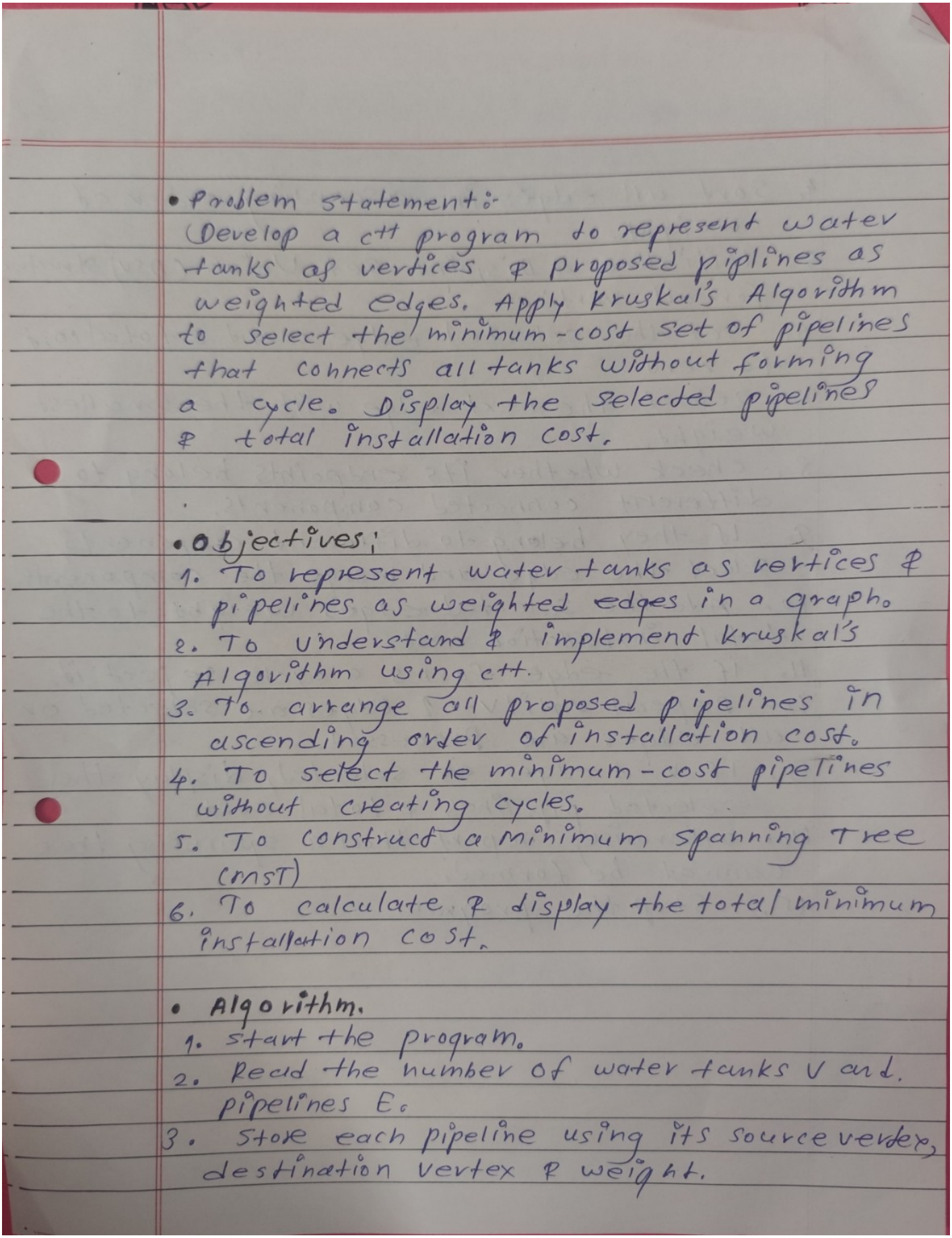
Kruskal’s Algorithm – Water Tank Pipeline MST

Problem Statement, Objectives, Algorithm and Flowchart (handwritten notes)

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• Problem statement:-
Develop a ctt program to represent water tanks as vertices & proposed piplines as weighted edges. Apply kruskal's Algorithm to select the minimum-cost set of pipelines that connects all tanks without forming a cycle. Display the selected pipelines & total installation cost.

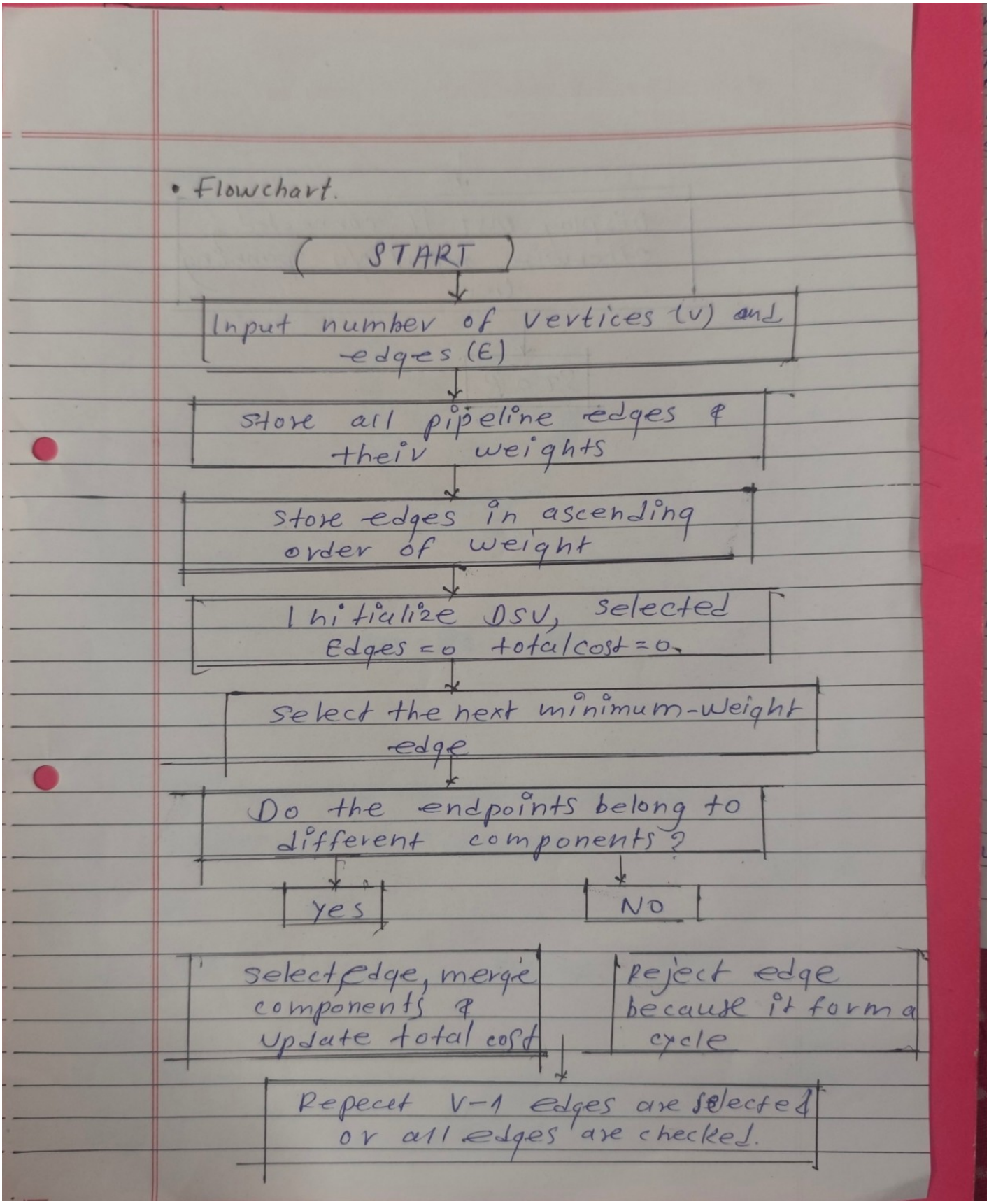
• Objectives;
1. To represent water tanks as vertices & pipelines as weighted edges in a graph.
2. To understand & implement kruskal's Algorithm using ctt.
3. To arrange all proposed pipelines in ascending order of installation cost.
4. To select the minimum-cost pipelines without creating cycles.
5. To construct a minimum spanning Tree (MST)
6. To calculate & display the total minimum installation cost.

• Algorithm.
1. start the program.
2. Read the number of water tanks V and pipelines E.
3. Store each pipeline using its source vertex, destination vertex & weight.

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4. Sort all edges in ascending order of weight.
5. Initialize a Disjoint Set Union (DSU) structure for all vertices.
6. set the selected edge count & total cost to zero
7. select the next edge with the smallest weight.
8. check whether its endpoints belong to different connected components.
9. If they belong to different components, select the edge and merge the components.
10. Add the selected edge's weight to the total installation cost.
11. If the edge forms a cycle, reject it.
12. Repeat until $V-1$ edges are selected or all edges are processed.
13. If $V-1$ edges are selected, display the selected pipelines & total cost.
14. otherwise, report that a spanning tree cannot be formed.
15. Stop the program.

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